

PROBLEM STATEMENT AND OBJECTIVES

1. Perform data preprocessing, cleaning, and exploratory data analysis (EDA). Analyze temporal patterns, trends, and seasonality in Global Active Power.
2. Conduct stationarity testing using the Augmented Dickey-Fuller (ADF) Test.
3. Develop and implement ARIMA models.
4. Apply SARIMA models to capture seasonal patterns in electricity consumption.
5. Build and evaluate deep learning models, including:
 - Recurrent Neural Networks (RNN)
 - Long Short-Term Memory (LSTM) Networks
6. Compare the forecasting performance of ARIMA, SARIMA, RNN, and LSTM models using appropriate evaluation metrics such as MAE, RMSE, and MAPE.
7. Generate forecasts of Global Active Power consumption for the next years and visualize future consumption trends.